Part 8: The MIT Bag Model and the Quark-Gluon Plasma

A Different Confinement Picture That Gets Us to Deconfinement

PHYS 7502 – Final Reading Assignment

1 A Different Picture of the Hadron

Everything we've developed in Parts 5-7 has been built around the *string* picture of confinement: the color field between two quarks is compressed into a flux tube, the tube has tension, and hadronization happens by string breaking. This is a beautiful picture, and it's what PYTHIA uses.

But it's not the only way to think about confinement. There's another picture – the **MIT bag model** – that starts from a different perspective and ends up making some complementary predictions, including a striking route to the **quark-gluon plasma** phase transition.

Complementary Pictures

It's useful to have two different mental models of confinement, each of which captures different physical aspects:

- **String picture**: confinement lives in the narrow color flux tube. Good for imagining separation and hadronization.
- **Bag picture**: confinement lives in a finite spatial region where quarks are free. Good for imagining a single hadron and its deconfinement.

These aren't competing theories; they're two approximations to the same underlying QCD. The bag model gets you to QGP thinking with relatively little effort.

2 The MIT Bag: Physical Picture

The MIT bag model was developed in 1974 at (surprise) MIT. The core idea is shockingly simple:

The Bag Model Assumptions

1. Inside a region of space called the **bag**, quarks move as *essentially free relativistic particles* (massless for simplicity).
2. Outside the bag, quarks are *infinitely massive* (can't propagate there at all).
3. The boundary of the bag is held in place by a phenomenological inward **bag pressure** B pressing against the outward kinetic pressure of the quarks.
4. For ground-state hadrons, the bag is *spherical*.

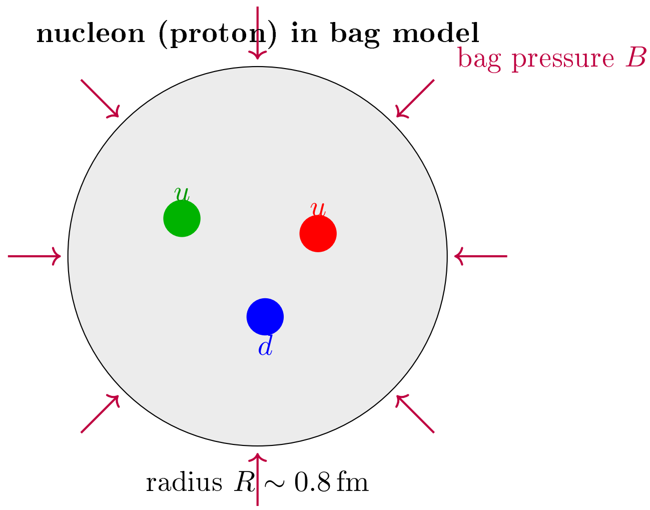


Figure 1: The MIT bag picture of a nucleon. Three quarks move as essentially free particles inside a spherical "bag". The boundary is held stable by an inward bag pressure B (a phenomenological parameter accounting for non-perturbative QCD vacuum effects) pushing against the outward kinetic pressure of the quarks.

This Model Is Crude

Griffiths is characteristically direct about this: *"No one would pretend that [the bag model] is a realistic picture of hadron structure."* It's a toy model. But it's an instructive one, and some of its predictions are quantitatively reasonable. For a model with just *one free parameter* (B), that's impressive.

3 Solving the Bag Model

Let's work through the calculation of a bag-model hadron. The setup: N massless quarks in a spherical bag of radius R.

3.1 Dirac equation for a massless quark in a sphere

Inside the bag, a massless quark satisfies the free Dirac equation:

$$\gamma^\mu \partial_\mu \psi = 0 \tag{1}$$

Separating into positive-energy modes $\psi = \psi_+ e^{-ip^0 t}$, with spinors built from two-component pieces $\psi = \binom{\psi_+}{\psi_-}$, the Dirac equation becomes:

$$\begin{pmatrix} p^0 & -\vec{\sigma}\cdot\vec{p} \\ +\vec{\sigma}\cdot\vec{p} & -p^0 \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = 0 \tag{2}$$

The lowest-energy (ground-state) solution is the $s_{1/2}$ mode, for which

$$\psi_+(\vec{r}, t) = \mathcal{N} e^{-ip^0 t} j_0(p^0 r) \chi_+ \tag{3}$$

where j_0 is the spherical Bessel function of order zero and χ_+ is a two-component spinor.

3.2 The crucial boundary condition

The bag model's defining feature is the **confinement boundary condition**: the normal component of the quark current vanishes at the bag surface. This translates into the condition that the scalar density $\bar{\psi}\psi$ vanishes at $r = R$. Working this out:

$$[j_0(p^0 R)]^2 - [j_1(p^0 R)]^2 = 0 \tag{4}$$

The Bag Model Eigenvalue

From the tabulated spherical Bessel functions, the lowest root of the boundary condition is:

$$\boxed{p^0 R = 2.04} \tag{5}$$

So the ground-state quark energy is

$$\varepsilon_n = \frac{2.04}{R} \tag{6}$$

(in natural units). This is a purely kinematic consequence of confining a massless relativistic particle in a sphere of radius R.

3.3 Total energy of a bag with N quarks

Each quark has kinetic energy $\varepsilon = 2.04/R$. For N quarks:

$$E_{\text{kin}}(R) = N \cdot \frac{2.04}{R} = \frac{2.04N}{R} \tag{7}$$

The bag itself stores energy from the pressure: you have to do work against B to create a volume-V bag, so the potential energy is

$$E_{\text{pot}}(R) = B \cdot V = B \cdot \frac{4\pi}{3} R^3 \tag{8}$$

Total energy:

$$E(R) = \frac{2.04N}{R} + \frac{4\pi}{3} R^3 B \tag{9}$$

3.4 The equilibrium radius

Minimize $E(R)$ with respect to R:

$$\frac{dE}{dR} = -\frac{2.04N}{R^2} + 4\pi R^2 B = 0 \tag{10}$$

Solving for R:

$$R^4 = \frac{2.04N}{4\pi B} \quad \Longrightarrow \quad R = \left(\frac{2.04N}{4\pi B}\right)^{1/4} \tag{11}$$

The Physical Balance

This is just a pressure balance.

- **Quarks want a big bag** – bigger bag means longer wavelength means lower kinetic energy. The $1/R$ term in equation (9) pulls R up.

- **The vacuum wants a small bag** – more bag means more pressure work, more energy. The R^3 term pulls R down.

The equilibrium radius is where these two pressures balance. It's a genuinely quantum-mechanical equilibrium: the $1/R$ scaling of the kinetic term comes from the uncertainty principle.

3.5 Fixing the bag constant

Equation (11) gives R as a function of the one free parameter B. To pin down B, we fit to some measured hadron property. The traditional choice: the nucleon mass.

For a nucleon ($N = 3$), plug in $R = 0.8\,\text{fm}$ (measured) to equation (11) and solve for B:

The Bag Constant

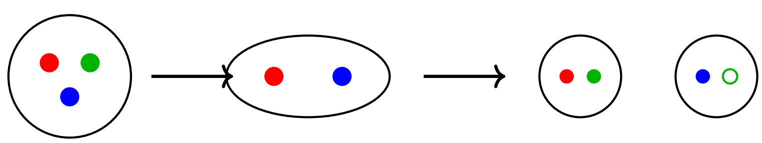
$$B^{1/4} \approx 206\,\text{MeV} \tag{12}$$

(Different fits give values in the range 145 MeV to 235 MeV.)

With this value of B, we can then *predict* other hadron properties – masses of excited states, radii of other baryons, etc. Some predictions work well, some are off by 20-30%. For a model with one parameter, this is not bad.

4 The Hadronization Picture in the Bag Model

Here's where the bag model gives you a physically beautiful qualitative picture of confinement and hadronization:



(a) quarks free inside (b) stretch to pull quarks apart (c) bag splits into two new bags (mesons/baryons)

Figure 2: The hadronization picture in the bag model. (a) Quarks move freely inside the original bag. (b) Trying to pull a quark out requires stretching the bag, at roughly 1 GeV/ fm. (c) When the stretching energy exceeds the $q\bar{q}$ pair production threshold, new quarks appear and the bag splits. Each smaller bag is a hadron.

This is the *same* pair-production-breaks-the-confinement story as the string model, but it's told in the bag language. Try to pull a quark out: the bag stretches, stores energy at $\sim 1\,\text{GeV/fm}$, until the stored energy exceeds $2m_q$ and a new pair pops out, closing off a new bag.

Bag vs. String: Same Physics, Different Picture

Both models predict the linear confinement potential $V(r) = \kappa r$ with $\kappa \sim 1\,\mathrm{GeV/fm}$ for quarks separated beyond the hadronic scale. Both predict that trying to isolate a quark produces a new hadron instead, via pair production. The difference is just in how you picture the *confining medium*: a 1D string (string model) vs. a 3D bubble (bag model). For static properties of hadrons, the bag picture is often more convenient.

5 The Path to the Quark-Gluon Plasma

Here's where the bag model really earns its keep. The same model that describes a single hadron also predicts a **phase transition** to a deconfined state of matter at high temperature or density.

5.1 The thermodynamic argument

Imagine you take a hot, dense collection of hadrons – say, the matter formed in a heavy-ion collision. Each hadron is a bag of quarks.

Now raise the temperature. Inside each hadron, the quarks have more kinetic energy, so they push outward on the bag more strongly. The bag walls have to push back with constant pressure B.

The Deconfinement Condition

The bag is stable only as long as the inward bag pressure B exceeds the outward kinetic pressure of the quark (and gluon) gas inside. If at some temperature T_c the quark-gluon gas pressure exceeds B:

$$P_{\mathrm{QG}}(T_c) > B \tag{13}$$

then the bag can no longer confine – it blows open, and the quarks and gluons spill out as a **deconfined plasma**.

5.2 Computing T_c

The pressure of a relativistic gas of massless particles is

$$P = g_{\mathrm{total}} \cdot \frac{\pi^2}{90} T^4 \tag{14}$$

where g_{total} is the total number of degrees of freedom (summing fermions with a factor 7/8 to account for Fermi-Dirac statistics).

For a QGP with 2 light flavors:

- **Gluons**: $g_g = N_c^2 - 1 \times 2_{\mathrm{polarizations}} = 8 \cdot 2 = 16$.
- **Quarks**: $g_q = N_c \cdot N_s \cdot N_f = 3 \cdot 2 \cdot 2 = 12$.
- **Antiquarks**: $g_{\bar{q}} = 12$ (same as quarks).

Total:

$$g_{\mathrm{total}} = g_g + \frac{7}{8}(g_q + g_{\bar{q}}) = 16 + \frac{7}{8} \cdot 24 = 16 + 21 = 37 \tag{15}$$

Setting $P_{\mathrm{QG}} = B$:

$$37 \cdot \frac{\pi^2}{90} T_c^4 = B \tag{16}$$

Solving for T_c:

QCD Critical Temperature from Bag Model

$$\boxed{T_c = \left(\frac{90}{37\pi^2}\right)^{1/4} B^{1/4}} \tag{17}$$

For $B^{1/4} = 206\,\mathrm{MeV}$, this gives

$$T_c \approx 144\,\mathrm{MeV} \tag{18}$$

This is remarkable. The bag model – with a single parameter B fit to the *zero-temperature* nucleon mass – predicts a quark-gluon plasma phase transition at around 140 MeV.

Comparison to Lattice QCD

Modern lattice QCD calculations find the QCD pseudo-critical temperature to be $T_c \approx 156\,\mathrm{MeV}$. The bag model gets this within about 10%! This is an extraordinary success for a one-parameter model whose main physical input was the nucleon mass.

5.3 The other axis: high density

A similar argument applies at high baryon density (e.g., inside neutron stars). Instead of a thermal gas, you have a cold but dense quark matter. The Fermi pressure of this matter can again exceed B.

The calculation gives a critical Fermi momentum of about 434 MeV, which corresponds to a baryon number density about **5 times nuclear saturation density**. If the interior of a neutron star is denser than this, it may contain quark matter rather than nucleons. This is an active area of astrophysical research.

5.4 The QCD phase diagram

Putting the two axes together produces the **QCD phase diagram**:

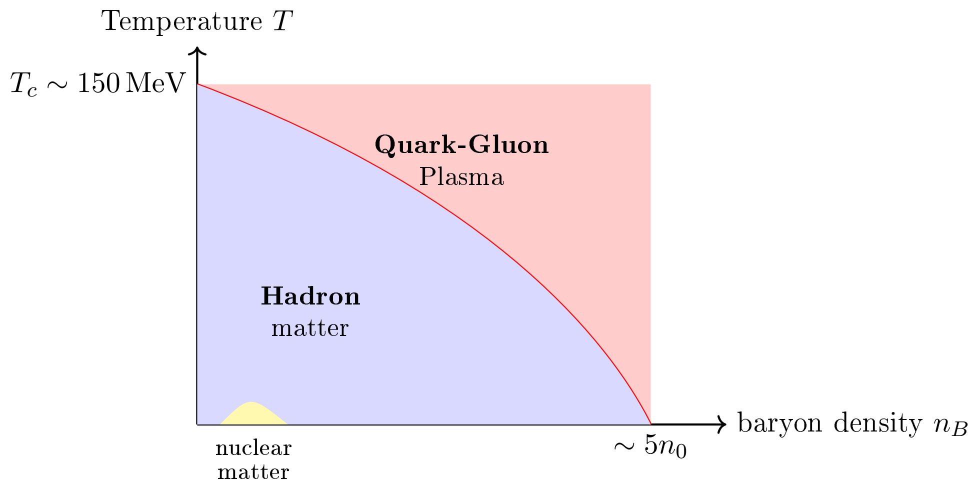


Figure 3: Schematic QCD phase diagram. At low T and low n_B, matter exists as a gas of hadrons. Ordinary nuclear matter sits near $T = 0$, $n_B = n_0$. At high T or n_B, the bag pressure is overcome and we enter the QGP phase. The boundary is the region of active experimental study.

6 Where QGP Lives Experimentally

6.1 Heavy-ion collisions

The experimental route to QGP is **relativistic heavy-ion collisions**: collide lead (Pb) with lead, or gold (Au) with gold, at the highest possible energies. At RHIC and the LHC, such collisions reach temperatures $T \sim 300\,\mathrm{MeV}$ – well above T_c.

The resulting fireball has been studied extensively, and many signatures of QGP formation have been observed:

- **Jet quenching**: high-p_T partons lose energy traveling through the medium.
- **Elliptic flow** v_2: the spatial asymmetry of the initial collision develops into a momentum asymmetry consistent with nearly perfect fluid behavior.
- **Strangeness enhancement**: more strange hadrons per pion than in pp collisions.
- J/ψ **suppression and recombination**.

6.2 Neutron star cores

Neutron stars have central densities that may reach $\sim 10n_0$, beyond the expected deconfinement threshold. Recent pulsar timing data (e.g., the very precise masses of NICER pulsars like J0740+6620) are starting to constrain the equation of state of ultra-dense matter.

7 Summary of Part 8

Main Takeaways

1. The **MIT bag model** treats hadrons as finite spherical regions where massless quarks move freely, bounded by an inward bag pressure B from the QCD vacuum.
2. Solving the confinement boundary condition gives $p^0 R = 2.04$, so the ground-state quark energy is $\varepsilon = 2.04/R$.
3. Minimizing total energy vs. R and fitting to the nucleon mass gives the one free parameter: $B^{1/4} \approx 206\,\mathrm{MeV}$.
4. Pulling a quark out stretches the bag; when stretching energy exceeds the $q\bar{q}$ pair threshold, the bag splits, producing new hadrons. Same hadronization story as in the string model, different picture.
5. When the internal quark/gluon pressure exceeds B (at high T or high density), the bag breaks open and we enter the **quark-gluon plasma** phase.
6. Setting $P_{\mathrm{QGP}} = B$ gives a critical temperature $T_c = (90/37\pi^2)^{1/4} B^{1/4} \approx 144\,\mathrm{MeV}$ – within 10% of the modern lattice-QCD value.
7. The bag model also predicts a critical baryon density of ~ 5 times nuclear saturation density. Neutron star cores may be above this threshold.
8. Heavy-ion collisions at RHIC and the LHC routinely create QGP, and its signatures (jet quenching, flow, strangeness enhancement) are observed and studied in detail.

8 Series Summary

Congratulations! You've made it through all eight parts. Here's the big-picture view of what you've just learned, organized by theme:

The Whole Story in One Page

The central problem was to understand hadrons and hadronization – the non-perturbative rearrangement of colored partons into colorless hadrons – in a regime where first-principles QCD is untractable.

The experimental route (Parts 2-3): jets are how we see partons in detectors. Sequential recombination algorithms (anti-k_T above all) are IRC-safe and give us the right bridge. Event generators (PYTHIA + shower + hadronization) simulate the whole process.

Factorization (Part 4): in any factorizable process, non-perturbative physics lives inside PDFs and fragmentation functions $D_i^h(z)$. Both evolve via DGLAP. FFs satisfy their own scaling violations, and MLLA predicts the hump-backed plateau.

The string picture (Parts 5-6): gluon self-interaction compresses color fields into flux tubes with tension $\kappa \approx 1\,\mathrm{GeV/fm}$. Schwinger pair production breaks the tube. The Lund model packages this into a concrete splitting function that PYTHIA uses.

The soft-scattering framework (Part 7): Regge theory plus the Pomeron (a trajectory with vacuum quantum numbers and $\alpha_0 \approx 1$) explains the high-energy near-constant total cross section. Dual Parton Model extends this to multi-particle production.
The bag picture (Part 8): a complementary view of confinement that leads beautifully into the quark-gluon plasma phase transition at $T_c \sim 150\,\mathrm{MeV}$.
The unifying thread: the string tension $\kappa \approx 1\,\mathrm{GeV/\,fm}$ appears in nearly every chapter. It's the string tension in Part 5, the Regge slope $\alpha' = 1/(2\pi\kappa)$ in Part 6, the Pomeron indirectly through the hadron spectrum in Part 7, and the bag pressure $B^{1/4} \approx 206\,\mathrm{MeV}$ shares the same energy scale in Part 8. One number, many manifestations.

You now have the conceptual toolbox to read modern particle physics papers on hadronization, jets, and soft QCD. Every framework you've met – PDFs and FFs, Lund strings, Pomerons, bag models, QGP – has a specific domain of applicability, and knowing which to reach for is a large part of being a practicing particle physicist.